

ECEN-601: Homework 6

Due on November 18, 2025 at 11:59 pm

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Problem 1

Consider $\mathbb{R}^4$ with the standard inner product. Let W be the subspace of $\mathbb{R}^4$ consisting of all vectors which are orthogonal to both $\alpha = (1, 0, -1, 1)$ and $\beta = (2, 3, -1, 2)$. Find a basis for W .

Solution

Constructing a matrix from these vectors:

$$\begin{bmatrix} 1 & 0 & -1 & 1 \\ 2 & 3 & -1 & 2 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & -1 & 1 \\ 0 & 3 & 1 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & -1 & 1 \\ 0 & 1 & 1/3 & 0 \end{bmatrix}$$

From these pivot values, $x_1 = x_3 - x_4$ and $3x_2 = -x_3$.

$$\begin{bmatrix} x_3 - x_4 \\ -x_3/3 \\ x_3 \\ x_4 \end{bmatrix} \rightarrow x_3 \begin{bmatrix} 1 \\ -1/3 \\ 1 \\ 0 \end{bmatrix} + x_4 \begin{bmatrix} -1 \\ 0 \\ 0 \\ 1 \end{bmatrix}$$

Thus, a basis for W is

$$\left\{ \begin{bmatrix} 1 \\ -1/3 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ 0 \\ 0 \\ 1 \end{bmatrix} \right\}$$

Problem 2

Let V be an inner product space, and let α and β be vectors in V . Show that $\alpha = \beta$ if and only if $\langle \alpha | \gamma \rangle = \langle \beta | \gamma \rangle$ for every γ in V .

Solution

By the definition of inner product, we know $\langle u + v, w \rangle = \langle u, w \rangle + \langle v, w \rangle$. Then, if we have $\alpha = \beta$, then

$$\begin{aligned}\langle \alpha | \gamma \rangle = \langle \beta | \gamma \rangle &\implies \langle \alpha | \gamma \rangle - \langle \beta | \gamma \rangle = 0 \\ &\qquad \qquad \qquad \langle \alpha - \beta | \gamma \rangle = 0\end{aligned}$$

Inner product orthogonality requires $\gamma = \alpha - \beta$, and since 0-equivalence means two vectors are equivalent, $\alpha = \beta$.

Problem 3

Let

$$\mathbf{p}_1 = \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix} \quad \mathbf{p}_2 = \begin{bmatrix} 4 \\ -2 \\ -6 \\ -7 \end{bmatrix} \quad \mathbf{p}_3 = \begin{bmatrix} 3 \\ 4 \\ -2 \\ 1 \end{bmatrix}$$

and

$$\mathbf{x} = \begin{bmatrix} 1 \\ 2 \\ 3 \\ 7 \end{bmatrix}.$$

Determine the best approximation, $\hat{\mathbf{x}}$, of the vector $\mathbf{x}$ by vectors in $W = \text{span}\{\mathbf{p}_1, \mathbf{p}_2, \mathbf{p}_3\}$. Also, determine the projection of $\mathbf{x}$ onto the orthogonal complement of W .

Solution

We have matrix A

$$\begin{bmatrix} 1 & 4 & 3 \\ 2 & -2 & 4 \\ 3 & -6 & -2 \\ 4 & -7 & 1 \end{bmatrix}$$

Finding the Gramian matrix, G (also $A^H A$):

$$\begin{bmatrix} \langle \mathbf{p}_1, \mathbf{p}_1 \rangle & \langle \mathbf{p}_1, \mathbf{p}_2 \rangle & \langle \mathbf{p}_1, \mathbf{p}_3 \rangle \\ \langle \mathbf{p}_2, \mathbf{p}_1 \rangle & \langle \mathbf{p}_2, \mathbf{p}_2 \rangle & \langle \mathbf{p}_2, \mathbf{p}_3 \rangle \\ \langle \mathbf{p}_3, \mathbf{p}_1 \rangle & \langle \mathbf{p}_3, \mathbf{p}_2 \rangle & \langle \mathbf{p}_3, \mathbf{p}_3 \rangle \end{bmatrix} = \begin{bmatrix} 30 & -46 & 9 \\ -46 & 105 & 9 \\ 9 & 9 & 30 \end{bmatrix}$$

Then the best approximation is $\hat{\mathbf{x}} = AG^{-1}A^T \mathbf{x}$.

$$\hat{\mathbf{x}} = \begin{bmatrix} 1 & 4 & 3 \\ 2 & -2 & 4 \\ 3 & -6 & -2 \\ 4 & -7 & 1 \end{bmatrix} \begin{bmatrix} 30 & -46 & 9 \\ -46 & 105 & 9 \\ 9 & 9 & 30 \end{bmatrix}^{-1} \begin{bmatrix} 1 & 4 & 3 \\ 2 & -2 & 4 \\ 3 & -6 & -2 \\ 4 & -7 & 1 \end{bmatrix}^T \begin{bmatrix} 1 \\ 2 \\ 3 \\ 7 \end{bmatrix}$$

By WolframAlpha, $\hat{\mathbf{x}} \approx (0.89, 2.97, 4.12, 5.70)^T$.

Problem 4

Let S be a subset of an inner product space V . Show that $(S^\perp)^\perp$ contains the subspace spanned by S . When V is finite-dimensional, show that $(S^\perp)^\perp$ is the subspace spanned by S .

Solution

We know that $S^\perp = \{\underline{v} \in V \mid \langle \underline{v} \mid \underline{s} \rangle = 0, \forall \underline{s} \in S\}$. Then $(S^\perp)^\perp = \{\underline{v} \in V \mid \langle \underline{v} \mid \underline{w} \rangle = 0, \forall \underline{w} \in S^\perp\}$. By these definitions, every vector in S is orthogonal to every vector in $S^\perp$.

This is equivalent to saying $s \in (S^\perp)^\perp, \forall s \in S$. Thus, the span of S is contained in $(S^\perp)^\perp$.

Problem 5

Let V be a finite-dimensional inner product space, and let $\mathcal{B} = (\alpha_1, \dots, \alpha_n)$ be an orthonormal basis for V . Let T be a linear operator on V ; and A , the matrix of T in the ordered basis $\mathcal{B}$. Prove that

$$A_{ij} = \langle T\alpha_j \mid \alpha_i \rangle$$

Solution

With orthonormal basis $\mathcal{B} = (\alpha_1, \dots, \alpha_n)$, we have

$$\underline{v} = \sum_{k=1}^n \langle \underline{v} \mid \alpha_k \rangle \alpha_k$$

Applying the transformation shows us that

$$T\alpha_j = \sum_{k=1}^n A_{kj} \alpha_k$$

but since

$$T\alpha_j = \sum_{k=1}^n \langle T\alpha_j \mid \alpha_k \rangle \alpha_k$$

we have that

$$A_{ij} = \langle T\alpha_j \mid \alpha_i \rangle$$

Problem 6

Suppose $V = W_1 \oplus W_2$ and that f_1 and f_2 are inner products on W_1 and W_2 , respectively. Show that there is a unique inner product f on V such that

- (a) $W_2 = W_1^\perp$;
- (b) $f(\alpha, \beta) = f_k(\alpha, \beta)$, when α, β are in $W_k, k = 1, 2$.

Solution

With $V = W_1 \oplus W_2$, we have $\forall v \in V, v = w_1 + w_2$ with $w_1 \in W_1$ and $w_2 \in W_2$.

We have $f(\alpha_1, \beta_1) = f_1(\alpha_1, \beta_1)$ for $\alpha_1, \beta_1 \in W_1$ and $f(\alpha_2, \beta_2) = f_2(\alpha_2, \beta_2)$ for $\alpha_2, \beta_2 \in W_2$.

By linearity of inner products, we have

$$\begin{aligned} f(\alpha_1 + \alpha_2, \beta_1 + \beta_2) &= f(\alpha_1, \beta_1) + f(\alpha_1, \beta_2) + f(\alpha_2, \beta_1) + f(\alpha_2, \beta_2) \\ &= f_1(\alpha_1, \beta_1) + 0 + 0 + f_2(\alpha_2, \beta_2) \\ &= f_1(\alpha_1, \beta_1) + f_2(\alpha_2, \beta_2) \end{aligned}$$